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RESEARCH ARTICLE

Quantum circuits for word embeddings

Hiroshi Ohno[1]

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Abstract

We propose a quantum circuit for word embeddings, serving as a quantum counterpart to the classical Word2vec model. While there are some quantum-inspired models in the literature, no quantum circuits have been proposed for this purpose. The proposed quantum circuit, which we call Q-word2vec, consists of two components: an encoding part that maps data into a Hilbert space and a decoding part. These components are implemented using parameterized unitaries and entangling unitaries, which introduce quantum entanglement into the circuit. Given word-context pairs as training data, Q-word2vec is trained using a hybrid quantum-classical framework. The embedding data are then obtained by measuring the output states of the encoding part. A loss function that incorporates the correlation coefficients between the distance vectors of the embedding data and those of the labels in the training data is used to enhance the performance of Q-word2vec. Furthermore, a heuristic formula is proposed for estimating the circuit length. Synthetic data and two small corpora are used to compare the performance of Q-word2vec with that of Word2vec. Although the performance of Q-word2vec is slightly inferior to that of Word2vec, we believe that leveraging the Hilbert space makes Q-word2vec a promising model.

Keywords Word embeddings · Word2vec · Skip-gram model · Quantum language models · Quantum coding

1 Introduction

Quantum machine learning (QML) (Maria Schuld and Petruccione 2015; Biamonte et al. 2017; Benedetti et al. 2019) is rapidly developing for several applications, including classification problems (Havlíček et al. 2019), regression (Yu et al. 2024), and generative models (Dugan et al. 2023) across many fields addressed by classical machine learning (ML). A number of successful outcomes have been reported. Influenced by the results reported in previous studies (Abbas et al. 2021; Caro et al. 2022; Gili et al. 2024; Huang et al. 2021; Yu et al. 2024), many practitioners of quantum and classical machine learning (ML) aim to achieve quantum advantages such as faster training and improved generalization performance. For example, from a theoretical perspective, the quantum advantage in generalization performance has been clarified under certain conditions regarding the training data. Huang et al. (2021) proposed a projected quantum kernel model that provides a quantum speed-up for learning tasks and demonstrated significant predictive advantages over classical models on engineered datasets designed to showcase maximal quantum advantage. Yu et al. (2024) found that when the smoothness index of the target function is not too low, parameterized quantum circuits offer an advantage over nearly optimal deep neural networks with rectified linear units for the same function approximation tasks. Both studies provided theoretical guarantees for quantum advantage. However, we believe that the likelihood of data science or ML practitioners encountering the above-mentioned data or task settings is low. In this study, toward quantum advantage, we aim to leverage the exponentially large Hilbert space that grows with the number of qubits in a quantum system.

Recently, in the field of artificial intelligence (AI), large language models (LLMs) are steadily growing, as evidenced by the advent of ChatGPT, where word or image processing plays a central role. In quantum computing (or quantum machine learning, QML) for word processing, words need to be converted into quantum states in the Hilbert space. If this conversion can be achieved, then we may see in QML (or quantum AI) the possibility of utilizing a huge number of words in the Hilbert space.

✉ Hiroshi Ohno
oono-h@mosk.tytlabs.co.jp

[1] Toyota Central R&D Labs., Inc., 41-1 Yokomichi, Nagakute, Aichi 480-1192, Japan

There are some studies of language models using quantum computing. For example, Xie et al. (2015) proposed a quantum information retrieval (IR) model based on the equivalence between quantum entanglement and unconditional pure dependence (UPD) (Hou et al. 2013) in a post-measurement configuration. Here, UPD relates to capturing contextual associations between words, while quantum entanglement is regarded as representing word co-occurrences. Experimental results demonstrated its effectiveness compared to classical IR models. Zhang et al. (2018) proposed a quantum-inspired IR model, specifically a question-answering system, based on neural networks. Using word embedding data, they constructed two density matrices to represent a question and an answer, respectively, and then combined them into a joint density matrix. A convolutional neural network was employed to extract useful similarity patterns from the joint density matrix. Experimental results showed that the proposed model outperformed state-of-the-art models. For text classification tasks, Li et al. (2018) proposed a method for quantum-inspired word embeddings. In their approach, each word, represented as a one-hot vector, is mapped into a superposition state (a complex vector) in the Hilbert space. Using these complex embeddings, a density matrix is calculated to represent a sentence. To perform this calculation, two types of networks were proposed for the task: a complex embedding superposition network and a complex embedding mixture network. The proposed networks demonstrated superior performance compared to state-of-the-art classical models in binary sentence classification tasks.

In the previous studies, as described above, quantum-inspired approaches have been proposed; however, no quantum circuits for word embeddings have been developed. Therefore, in this study, we focus on designing quantum circuits for word embeddings.

We construct a quantum circuit for word embeddings, inspired by the (classical) Word2vec model (Mikolov et al. 2013), which is composed of neural networks. The quantum circuit (called Q-word2vec) follows the structure of Word2vec, comprising two parts: an encoding part and a decoding part. It consists of Pauli rotation gates (parameterized unitaries) and controlled Pauli-X gates (for providing entanglement), arranged in a layered structure with adjustable circuit depth. Given a one-hot vector as input for a word and an output vector representing the word's context, the circuit is trained using a stochastic gradient algorithm with the parameter shift rule. To enhance training performance, we introduce an additional loss term to the loss function, which incorporates the correlation coefficient between the distance vectors of the embedding data and those of the training labels. Using synthetic datasets, we investigate the performance of Q-word2vec on two-, three-, four-, and five-qubit quantum systems. Compared to the Word2vec baseline, Q-word2vec achieves approximately 70% accuracy. Using small corpora, we perform two-dimensional representations of word embeddings and word vector calculations for both Q-word2vec and Word2vec. The results indicate that Q-word2vec performs slightly worse than Word2vec. Although these results proved somewhat inferior to those of Word2vec, we are convinced that Q-word2vec holds significant promise.

Our contributions are the followings:

- We propose a quantum circuit model, Q-word2vec, for word embeddings, which, to the best of our knowledge, is the first quantum model of its kind. In Q-word2vec, parameterized unitaries are trained using a stochastic gradient algorithm.
- To optimize circuit design, we introduce a heuristic method for estimating circuit length, inspired by dense quantum coding (Ambainis et al. 2002). By leveraging this estimation, we can efficiently search for the optimal circuit length, thereby reducing search time and effort.
- We propose a new loss function to improve the performance of Q-word2vec. This function utilizes the correlation coefficient between the distance vectors of the embedding data and those of the labels of the training data.

The remainder of the paper is organized as follows: Sect. 2 describes Word2vec. In Sect. 3, Q-word2vec is introduced. In addition, the proposed loss function and a heuristic circuit length estimation are described. We define our synthetic data and the evaluation metrics in Sect. 4. Section 5 describes the experimental results for the synthetic datasets and small corpora. A summary of the experimental results are provided. Finally, Sect. 6 concludes the study including future work.

2 Classical Word2vec

We describe a classical Word2vec employed in this study. Word2vec is a type of shallow neural network with linear activation functions in the hidden and output layers. Hence, the neural network realizes a linear mapping between the input and output vectors. Corresponding to a word, in general, we prepare a one-hot vector as the input vector (i.e., a one-hot encoding). In this study, to develop Word2vec, we address the skip-gram model (Mikolov et al. 2013). In the skip-gram model, the output vector corresponding to an input vector represents the context of the word in the input vector. The context is represented by the surrounding words (in this study, the number of surrounding words is two). For example, in the sentence "New York City," the surrounding words corresponding to the word "York" are "New" and "City." For the training data, the element of the target (output) vector is

one for the surrounding words and zero otherwise. In this study, however, in comparison with Q-word2vec, we used 0.5 for the surrounding words.

Formally, Word2vec is constructed by the following neural network:

$$H_v = W^{HI} I_v, \tag{1}$$
$$O_v = W^{OH} H_v, \tag{2}$$

where I_v denotes the input vector, W^{HI} denotes the hidden-input matrix, H_v denotes the hidden (or embedding) vector, W^{OH} denotes the output-hidden matrix, and O_v denotes the output vector. For p words, the size of I_v and O_v is $p \times 1$. The estimated vector $\hat{O}_v$ based on the output vector is calculated by the soft-max function as follows:

$$\hat{o}_i = \frac{\exp(o_i)}{\sum_{j=1}^{p} \exp(o_j)}, \tag{3}$$

where o_i denotes the i-th element of the output vector.

For the network training, the loss (or cost) function uses the cross-entropy function (Bishop 2006) whose arguments are the estimated and target (ground truth) vectors. After training, embedding data composed of the embedding vectors are utilized for such tasks as word similarity measure. The trainability of Word2vec is mainly dominated by the non-linearity of the soft-max function. In addition, Word2vec has two characteristics encouraging the trainability: (1) among the elements in the output vector, constraints such as the squared sum of the elements is equal to one do not exist, and (2) the elements can take values from minus infinity to plus infinity, which means that the dynamic range of the model parameters is large.

3 Methods

In this section, we introduce Q-word2vec and its loss function. Additionally, for embedding data, the evaluation metrics are presented, and for estimating the quantum circuit length in Q-word2vec before the experiments, a heuristic formula is given to reduce the parameter search cost.

3.1 Q-word2vec

Q-word2vec we propose has a similar structure to Word2vec. Figure 1 shows the structure of Q-word2vec. In this figure, $\mathrm{U}(\theta_u)$ and $\mathrm{V}(\theta_v)$ are composed of two types of unitary matrices: parameterized unitary matrices and entangling unitary matrices U_{enta}. n denotes the system qubits and n_e ($1 \leq n_e \leq n$) denotes the embedding qubits. The input

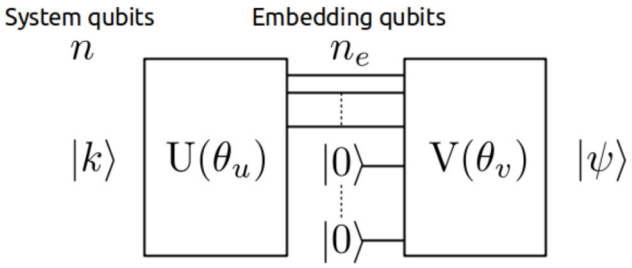


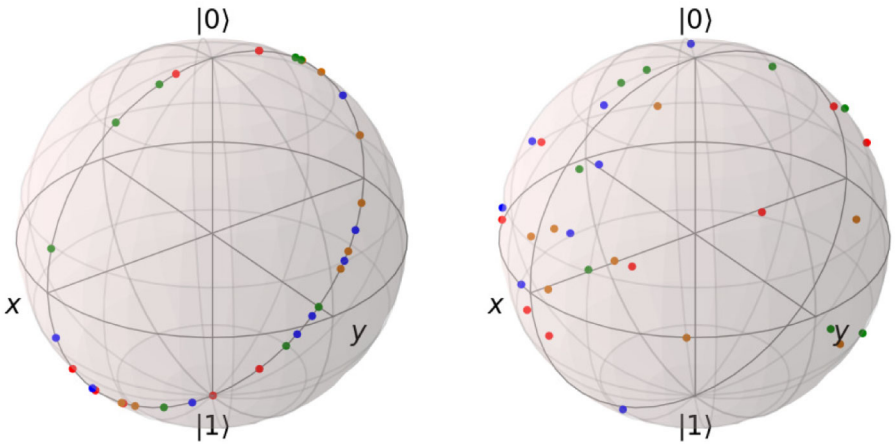
Fig. 1 Structure of Q-word2vec. $\mathrm{U}(\theta_u)$ and $\mathrm{V}(\theta_v)$ consist of parameterized unitary matrices, along with unitary matrices U_{enta} that introduce quantum entanglement

quantum state $|k\rangle$ corresponds to a one-hot vector (input vector) in Word2vec, where $0 \leq k \leq N-1$ ($N = 2^n$). We note that for the n-qubit system, the maximum training data size is N. Similar to the neural network for Word2vec, the linear mapping between $|k\rangle$ and $|\psi\rangle$ is realized by U and V.

The model ($\mathrm{U}(\theta_u)$ and $\mathrm{V}(\theta_v)$) uses the Pauli rotation matrices (R_X, R_Y, and R_Z) and unitary matrices (U_{enta}) to provide quantum entanglement, as follows:

$$\mathrm{U}(\theta) = \prod_{l=1}^{L} \mathrm{U}_{enta} R_Z(\theta_z^{L-l+1})^{\otimes n} R_Y(\theta_y^{L-l+1})^{\otimes n} R_X(\theta_x^{L-l+1})^{\otimes n}, \tag{4}$$

where L denotes the length of the quantum circuit. In Fig. 1, L_u and L_v refer to the lengths of the quantum circuits for U and V, respectively. Unless otherwise specified, L is equal to L_u or L_v. Note that in Eq. 4, the depth L (as well as L_u and L_v) does not include the depth of U_{enta}. The output quantum states $|\psi\rangle$ correspond to the context of the input words. The measurement of the output quantum states $|\psi\rangle$ is performed using the computational basis (Pauli Z basis). To compose U_{enta}, we used controlled Pauli X gates as the entangler and the circuit-block configuration (Sim et al. 2019) for creating the entanglement. From preliminary experiments comparing the controlled Pauli X and Z gates, the control Pauli X gates were selected. The entangling configuration affects the expressibility, entanglement capability, and trainability of the quantum circuit. According to Sim et al. (2019), the circuit-block configurations exhibited favorable expressibility and entanglement capability. Hence, we adopted the circuit-block configuration. Using the Pauli rotation matrices (R_X, R_Y, and R_Z), the words can be laid out throughout the Hilbert space. For example, in the case of one embedding qubit ($n_e = 1$), Fig. 2 shows the Bloch spheres for using one Pauli rotation matrix R_Y and three Pauli rotation matrices R_X, R_Y, and R_Z in $\mathrm{U}(\theta_u)$. In this figure, the number of words was 32 (i.e., $n = 5$). In Fig. 2a, the words are positioned on the circumference around the Y-axis, while in b, the words are



(a) (b)

Fig. 2 Words on Bloch spheres. **a** R_Y and **b** R_X, R_Y, and R_Z are used in $\mathrm{U}(\theta_u)$. The colored points correspond to the words. The number of points is 32

positioned over the entire sphere. In Eq. 4, we used three Pauli rotation matrices in Q-word2vec to secure a larger area for positioning words. To vectorize words in the Hilbert space, we measure the quantum states. In this study, we use a projective measurement in the computational basis. However, after the measurement, the relationships between words in the Hilbert space may not be preserved. Therefore, to retain these relationships, we modified the loss function as described below. For training Q-word2vec, we used a stochastic gradient method[1] using the parameter shift rule (Schuld et al. 2019) with the simultaneous perturbation stochastic approximation (SPSA) (Gacon et al. 2021). SPSA is an optimization technique where we randomly sample from the gradient in order to reduce the number of loss-function evaluations in the parameter shift rule. For the loss function, the cross-entropy loss function, which is the same as in Word2vec, is used, but to improve the quality of the embedding data, we modify the loss function as described below. Here, we note that the quality of the embedding data for Q-word2vec is assessed based on the embedding data of Word2vec.

3.2 Evaluation of embedding data

To evaluate the embedding data of Q-word2vec, we used the Pearson correlation coefficient between the distances based on the Q-word2vec and those of Word2vec data. After training, we measured the embedding qubit states of Q-word2vec and obtained the classical embedding vectors. One embedding vector corresponds to one word. Given embedding vectors generated by Q-word2vec or Word2vec, we construct a vector whose components are the Euclidean distances between these embedding vectors. We refer to this vector as the "distance vector." We then calculated the correlation coefficient between the distance vectors obtained from Word2vec and Q-word2vec. In this study, the distance vector derived from Word2vec is regarded as the ground truth. Ideally, Q-word2vec should achieve a higher correlation coefficient and generate high-quality embedding data.

The cross-entropy loss is not sufficient to evaluate the quality of the embedding data because, in general, quantum circuits with redundant parameters tend to lower the cross-entropy loss. Therefore, in addition to the cross-entropy loss, we utilized the correlation coefficient for the distance vectors for Word2vec and Q-word2vec.

3.3 Loss function for Q-word2vec

For Q-word2vec, the loss function (Loss) with cross-entropy loss includes an added term involving the correlation coefficient described in Sect. 3.2 as follows:

$$\begin{aligned}\mathrm{Loss} := \mathrm{CrossEntropyLoss} - C \cdot \mathrm{CorrelationCoefficient} \\ (EmbeddingData, TrainingLabels),\end{aligned} \tag{5}$$

where C (> 0) is a balancing parameter and CorrelationCoefficient($EmbeddingData$, $TrainingLabels$) denotes the correlation coefficient between the distance vector of the embedding data ($EmbeddingData$) in Q-word2vec measured at each parameter updating step and that of the labels in the training data ($TrainingLabels$). The reason for using the labels in the training data is that in the preliminary experiments, we observed a relatively high correlation between the distance vector of the embedding data of Word2vec and that of the labels in the training data (Appendix). By utilizing this correlation coefficient, we can

[1] The learning rate was set to 0.001 and the moment term was set to 0.9 in the experiments.

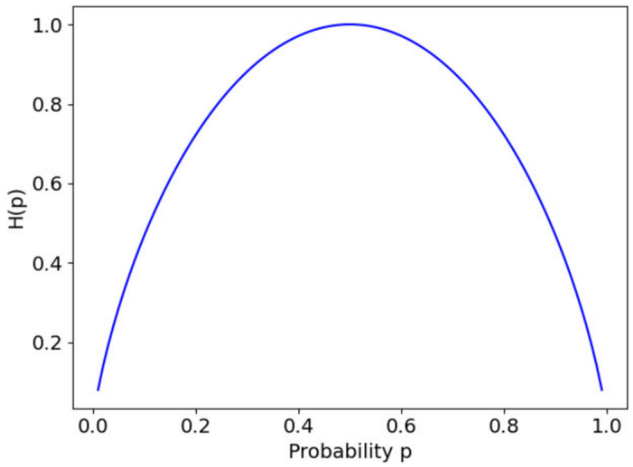


Fig. 3 Binary entropy $H(p)$. p denotes a probability referred to as error probability in this study

perform a hyper-parameter search for circuit length (L_u and L_v) and the balancing parameter (C) with respect to the quality of the embedding data.

3.4 Estimation of quantum circuit length

To reduce the burden of the preliminary experiments, such as a hyper-parameter search, estimating the model's parameters is desirable for practitioners. In this section, we propose a heuristic formula for estimating the length of a quantum circuit.

Given the number of training data ($\#data$), n, and n_e for Q-word2vec, we used the following heuristic formula for estimating L in Eq. 4:

$$L = \left\lceil \frac{\#data}{3(n+n_e)A_{th}} p(n,n_e) \right\rceil, \tag{6}$$

where A_{th} denotes a threshold value determined empirically and $p(n,n_e) \in [0,0.5)$ denotes the error probability based on n and n_e. The relationship among the error probability p and n, as well as n_e, is characterized by the inequality $n_e \geq (1-H(p))2^n$ (Ambainis et al. 2002; Aaronson 2007), where $H(p)$ denotes the binary entropy function as shown in Fig. 3.

Equation 6 is derived from the following inequality.

$$\frac{\#data}{3(n+n_e)L} p(n,n_e) \leq A_{th}. \tag{7}$$

In Eq. 7, the number 3 indicates three Pauli rotation matrices used. This inequality means that the average number of missing data per parameter which is passed through from U to V is smaller than or equal to the threshold A_{th}. Roughly speaking, A_{th} indicates the upper bounds of the number of missing data. In Eq. 7, $p(n,n_e)$ is estimated from the equality $n_e = (1-H(p))2^n$, given n and n_e.

4 Experiments

In the experiments, we used synthetic data and two small corpora consisting of sentence and word lists as training data. The synthetic data were constructed as follows: for the output vector of size N corresponding to $|k\rangle$, as two surrounding words (word context), two elements, randomly selected from the set $\{1,\ldots,N\}$, were set to 0.5, while all other elements were set to zero.

The cross-entropy loss does not exactly indicate whether the estimated output of Q-word2vec or Word2vec reproduces the training data labels (representing word contexts). Therefore, we introduce an error metric called the "error rate." The error rate is calculated as follows: it is the sum of the number of mismatches between the two peak positions (the two surrounding words) of the training data labels and the estimated model outputs, divided by twice the total number of training samples. In the experiments, we required that the error rate be zero for the training data.

The quantum circuits for Eq. 4 were implemented using the open-source software (Efthymiou et al. 2021).[2]

5 Results

In this section, we first explore the threshold value for the circuit length estimation and the effects of entanglement unitaries and circuit length in Q-word2vec. Second, we assess the quality of the embedding data generated by Q-word2vec for $n \in \{2,3,4,5\}$. Finally, we demonstrate applications for embedding data using the two small corpora.

5.1 Threshold value for circuit length estimation

To determine the C_{th} value, the following A values were plotted as shown in Fig. 4.

$$A = \frac{\#data}{3(n+n_e)L} p(n,n_e). \tag{8}$$

Here, the number of data points plotted was 45, which was selected from the range in $n \in \{2,3,4\}$, $1 \leq n_e \leq n$, and $L \in \{4,5\}$. In Fig. 4, the error (denoted by Error) corresponds to the error rate. Here, error one (Error = 1.0) indicates the results where the error rate is greater than zero.

[2] The software version used is as follows: Qibo 0.2.11. Word2vec was implemented using Pytorch (version 1.9.0).

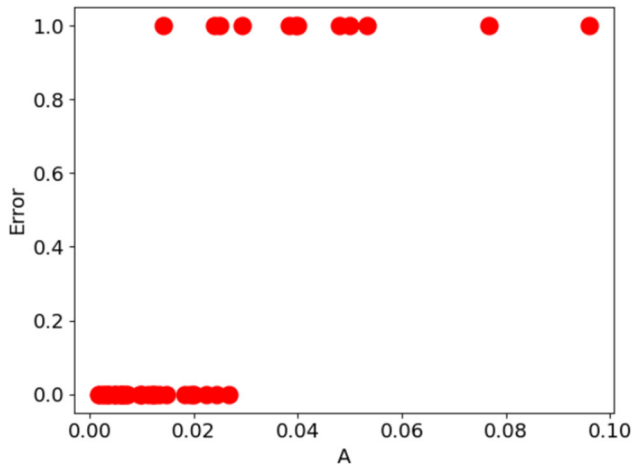


Fig. 4 Relation between error and A

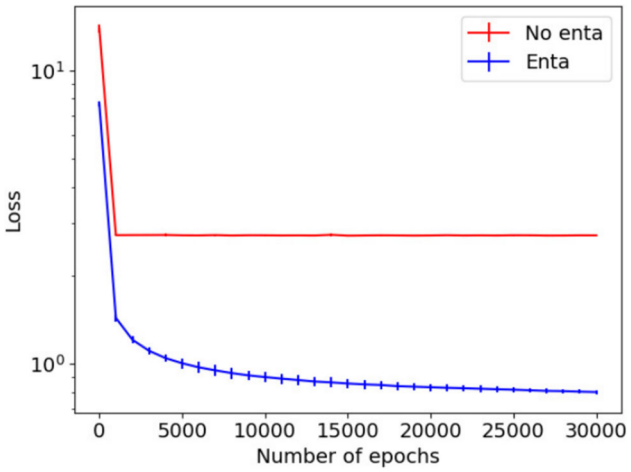


Fig. 5 Results of training for entanglement and no entanglement. For Q-word2vec, $n = 4$, $n_e = 3$, $C = 1$, and $L = 10$

From the figure, we set A_{th} to 0.02.

We applied the heuristic formula (Eq. 6) to the two cases: $n = 4, n_e = 1, C = 0, p = 0.35$, and $\#data = 16\ (= 2^n)$; and $n = 5, n_e = 3, C = 0, p = 0.32$, and $\#data = 32\ (= 2^n)$, where p denotes the error probability. We obtained $L = 19$ and $L = 22$ for these cases, respectively. However, after the training, the corresponding error rates were greater than zero. To achieve a zero error rate, we observed that the error probability p needs to be less than 0.3. For small data sets, however, there were cases where the error rate became zero: for example, $n = 5, n_e = 1, p = 0.40, \#data = 6\ (< 2^n)$, and $L = 7$. Exploration of the relationship between p and $\#data$ required to achieve a zero error rate is left for future work.

5.2 Effect of entanglement unitaries

The training curves for Q-word2vec, with and without entanglement unitaries U_{enta}, are shown in Fig. 5. Each training was conducted five times, with different random seeds.[3] The average cross-entropy loss and the standard deviation (indicated by the error bars) are shown.

At epoch zero, there was a significant difference in loss between the models with and without U_{enta}. It appears that the reduction in loss is primarily due to the entanglements. As the number of epochs increased, the loss with U_{enta} was gradually decreased. Therefore, we conclude that entanglements play an important role in Q-word2vec training. Note that, due to entanglement, the embedded words are located inside the Bloch sphere for a single qubit, thereby enhancing expressibility.

[3] The hyper-parameter values and training conditions are as follows: $n = 4, n_e = 3, C = 1, L = 10$, batch size = 8, and the max epoch was 30,000. The initial parameter values were set to zero.

5.3 Effect of circuit lengths of Q-word2vec

To assess the effect of circuit length L_u and L_v on the quality of the embedding data of Q-word2vec, we examined the correlation coefficients between the distance vectors of the embedding data of Word2vec and those of Q-word2vec. The maximum L_u and L_v were set to L estimated by the heuristic formula described in the previous section. For $n = 5$, the maximum values of L_u and L_v were set to 20, and n_e was set to 4 in order to reduce the error rate. Figure 6 shows the results of the correlation coefficient for $n = 2$, $n = 4$, and $n = 5$.[4]

For $n = 2$, the maximum correlation coefficient was 0.84, where $L_u = 5$ and $L_v = 2$. However, the error rate was 0.375. For the case where the error rate was zero, the maximum correlation coefficient for $L_u = 4$ and $L_v = 5$ was 0.55. For the cases where the error rate was zero, we found that $L_v \geq 3$. For $n = 4$, the maximum correlation coefficient was 0.22, where $L_u = 10$ and $L_v = 8$, and the error rate was zero. For $n = 5$, the maximum correlation coefficient was 0.14, where $L_u = 12$ and $L_v = 12$, and the error rate was zero. For many cases where $L_v > 6$, we found that the error rate was zero. Therefore, we should primarily focus on L_v for the unitary V to enhance the correlation coefficients.

5.4 Comparison of embedding data quality of Q-word2vec and Word2vec

In this section, we examined the quality of the embedding data of Q-word2vec comparing with that of Word2vec.

[4] For $n = 2, n_e = 1, C = 0$, and $\#data = 4$, batch size = 4, and the maximum epoch was 20,000 and for $n = 4, n_e = 3, C = 0$, and $\#data = 16$, batch size = 8, and the maximum epoch was 30,000. For $n = 5, n_e = 4, C = 0$, and $\#data = 32$, batch size = 16, and the maximum epoch was 30,000.

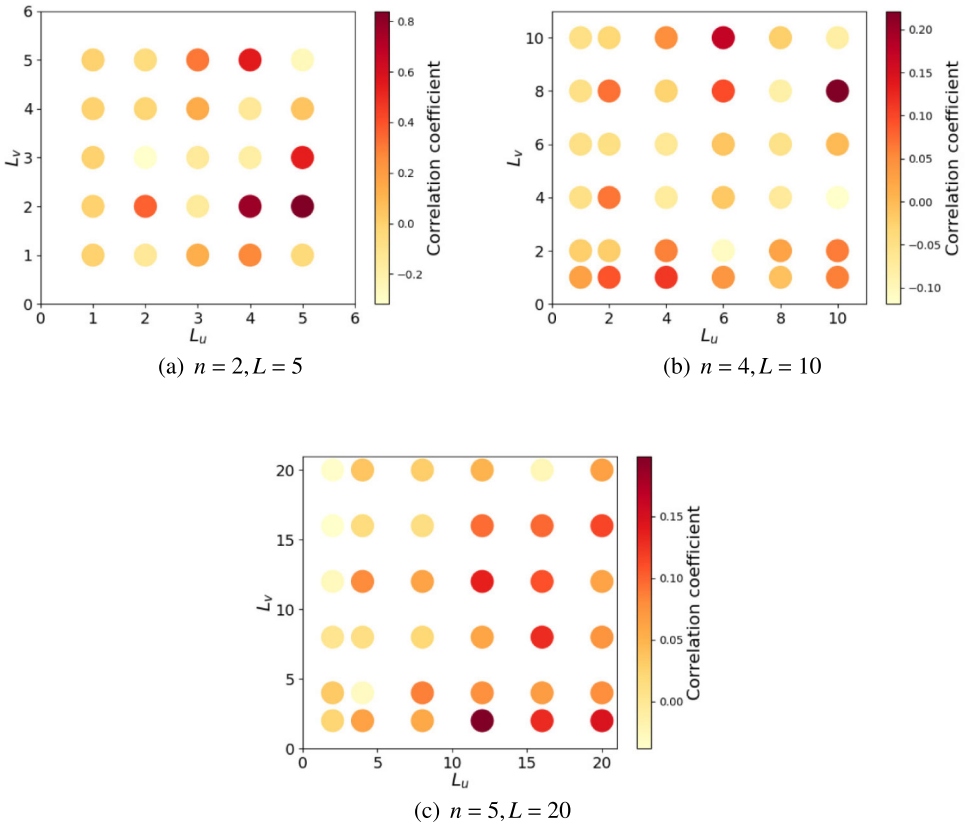


Fig. 6 Results of correlation coefficients for two-, four-, and five-qubit systems with varying circuit length L_u and L_v in Q-word2vec

For two-, three-, four-, and five-qubit systems[5], the corresponding results are presented in Fig. 7. In this figure, C-Q represents the correlation coefficients between the distance vectors generated by Word2vec and those generated by Q-word2vec, whereas C-C indicates the correlation coefficients between the distance vectors generated solely by Word2vec. For the label "C-Q (CC)," CC in parentheses denotes the use of the proposed cost function. Each box-plot is based on 25 data points.[6]

As shown in Fig. 7, increasing the number of qubits leads to a decrease in correlation coefficients across all cases. This trend may be attributed to the fact that distance vectors become large when the amount of training data increases. Additionally, the C-Q results exhibited greater variance compared to the C-C results. This suggests that Q-word2vec training is relatively sensitive to differences in random seed initialization. From the results of C-Q (CC) in Fig. 7, it is evident that the term Correlation-Coefficient(*EmbeddingData, TrainingLabels*) in Eq. 5 contributed to an improvement in the correlation coefficients. However, the performance of C-Q (CC) did not reach that of C-C; overall, the correlation coefficients were slightly less than 70% of those in the C-C case. As indicated in the definition of the loss function (Eq. 5), the correlation coefficient for C-Q (CC) is upper bounded by the correlation between the training labels and the Word2vec embedding data. We therefore believe that this upper-bound affected the results, which may be considered a limitation of the proposed loss function.

5.5 Applications for embedding data

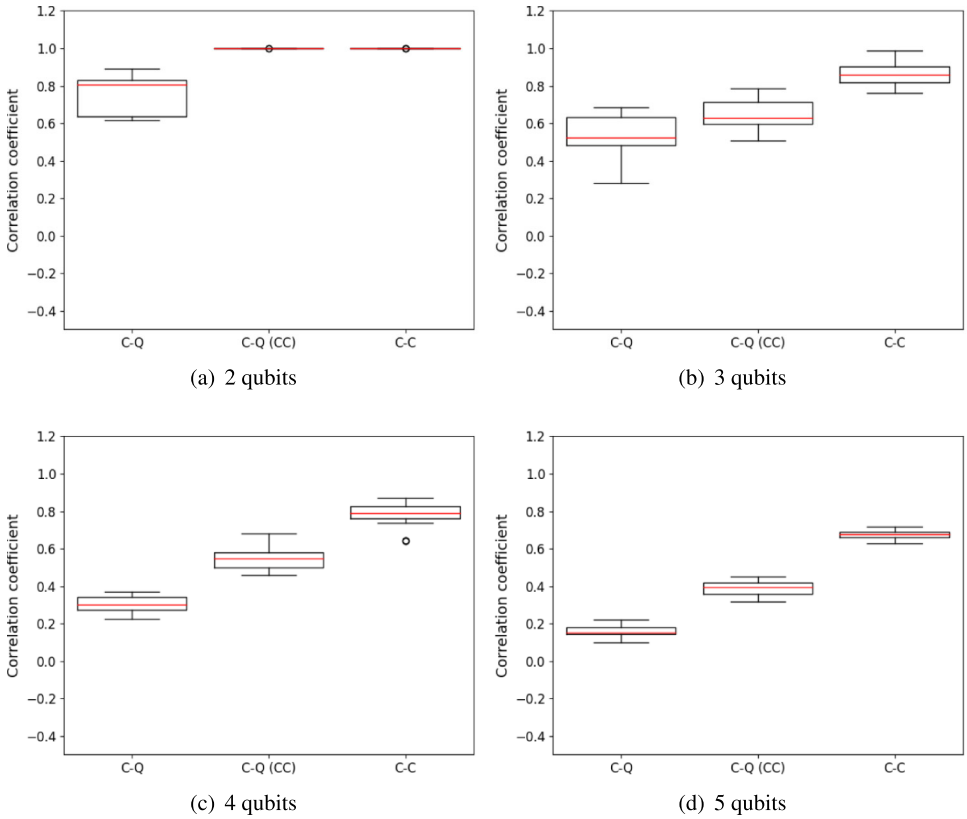
In this section, as applications for embedding data using small corpora, we addressed word embeddings and word vector calculations to demonstrate the usage of those data.

5.5.1 Word embeddings

We applied both the Q-word2vec and Word2vec models to the following list of sentences as a corpus: ["i like dog," "i like cat," "i like animal," "dog cat animal," "apple cat

[5] We performed the hyper-parameter search for L_v and C using the heuristic formula (Eq. 6) and the correlation coefficient values in the proposed loss function (Eq. 5). The hyper-parameter values and training conditions for each system are as follows: For two-qubit system, $n_e = 1$, $L_u = 5$, $L_v = 5$, $C = 1.5$, batch size = 2, and the maximum epoch was 20,000. For three-qubit system, $n_e = 2$, $L_u = 6$, $L_v = 6$, $C = 3.0$, batch size = 4, and the maximum epoch was 20,000. For four-qubit system, $n_e = 3$, $L_u = 10$, $L_v = 7$, $C = 1.0$, batch size = 8, and the maximum epoch was 30,000. For five-qubit system, $n_e = 4$, $L_u = 18$, $L_v = 14$, $C = 4.0$, batch size = 16, and the maximum epoch was 30,000.

[6] For example, for the case of "C-Q," the number of the distance vectors generated from Word2vec or Q-word2vec was five, and the number of their combinations was 25.



(a) 2 qubits (b) 3 qubits

(c) 4 qubits (d) 5 qubits

Fig. 7 Results of correlation coefficients between the distance vectors obtained from Word2vec and those of Q-word2vec for $n \in \{2, 3, 4, 5\}$

dog like," "dog fish milk like," "dog cat eyes like," "i like apple," "apple i hate," "apple i movie," "book music like," "cat dog hate," "cat dog like"].[7] From the list, we obtained 13 words and the corresponding context consisting of two words, resulting in a total of 13 training samples. We applied both the Q-word2vec model with the proposed loss function and the Word2vec model to the data. For Q-word2vec, the number of qubits was set to four ($n = 4$). We set the number of embedding qubits to two ($n_e = 2$), given that the error probability ($p = 0.295$) was below 0.3, and L was 11. After training,[8] the obtained word embeddings are plotted in Fig. 8. For Q-word2vec, the embedding data were reduced to two dimensions using the principal component analysis (PCA). The correlation coefficient for the distance vectors was 0.81; the error rate for both models was zero.

Figure 8a shows the Word2vec results, where the words "i" and "animal" form one group, while the words "dog" and "cat" form another group. Furthermore, the words "hate" and "movie" can be considered a group. Similar groupings are observed in Fig. 8b, which shows the Q-word2vec results.

5.5.2 Word vector calculation

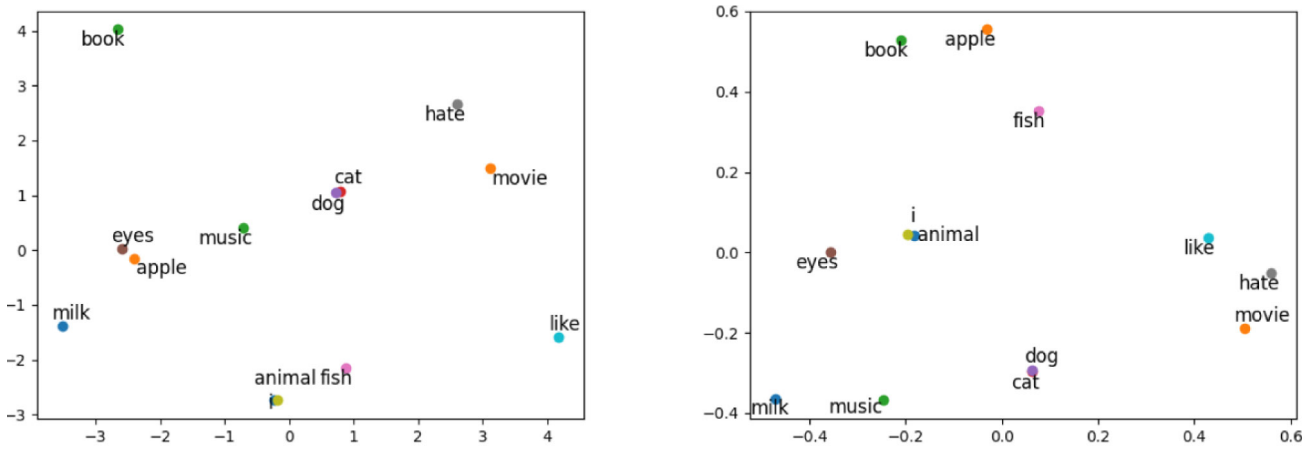
In the word vector calculations using embedding data, we investigated the addition and subtraction operations between embedding vectors. For capital and country words, we used the following list: ["capital," "beijing," "china," "capital," "tokyo," "japan," "capital," "moscow," "russia," "capital," "kinshasa," "congo," "capital," "jakarta," "indonesia," "capital," "lima," "peru," "capital," "cairo," "egypt," "capital"]. Then, the number of training data was 15. For Q-word2vec, the number of qubits was set to four ($n = 4$) and the number of embedding qubits was set to two ($n_e = 2$), and L was 13. After training,[9] we estimated a vector ($v_{country_i}$) corresponding to the country i using the following equations:

$$v_{i,j} = v_{country_j} - v_{city_j} + v_{city_i}, \tag{9}$$

[7] We obtained the sentences from the following site: https://towardsdatascience.com/skip-gram-neural-network-from-scratch-485f2e688238

[8] For the Word2vec training, the maximum epoch was 30,000; the batch size was 13. The number of hidden units was two (corresponding to the embedding data dimensionality). For the Q-word2vec training, the maximum epoch was 20,000, the batch size was 13, and C was set to 2.0.

[9] For the Word2vec training, the maximum epoch was 30,000; the batch size was 15. The number of hidden units was two. For the Q-word2vec training, the maximum epoch was 20,000, the batch size was 15, and C was set to 3.0.



(a) Word2vec (b) Q-word2vec

Fig. 8 Results of two-dimensional representation of word embeddings using PCA for Word2vec and Q-word2vec. The number of words was 13

and

$$\hat{v}_{i,j} = \arg\min_{v \in S_v} \|v_{i,j} - v\|, \tag{10}$$

where $\hat{v}_{i,j}$ denotes an estimated vector for $v_{country_i}$ and S_v denotes the overall vector set. The subtraction $v_{country_j} - v_{city_j}$ represents the context with respect to capital. For evaluation, we counted the number of correct answers using the following equation:

$$\mathrm{I}(v_{country_i} = \hat{v}_{i,j}), \tag{11}$$

where I denotes the indicator function.

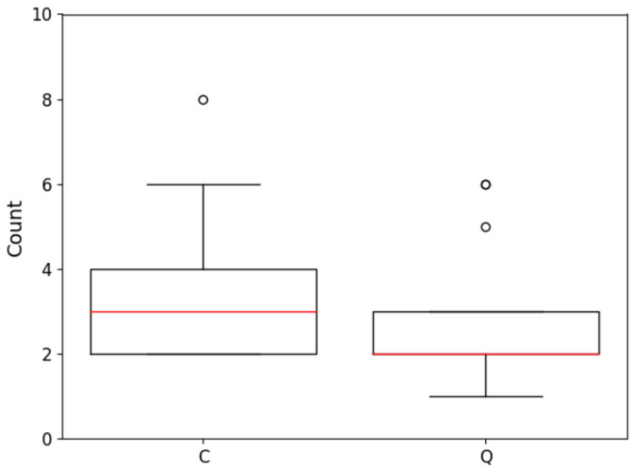


Fig. 9 Results of word vector calculations for country name estimation for Word2vec (C) and Q-word2vec (Q). The Y-axis is the number of correct answers. The number of words was 15

The results for Word2vec and Q-word2vec (respectively denoted by C and Q) are shown in Fig. 9. The number of data points for the box-plots was 20. The number of correct answers for Q-word2vec was about 80% of that for Word2vec on average.

5.6 Summary of the experimental results

In this section, we summarize the experimental results for both the synthetic data and the small corpora.

- Entanglements played an important role in the Q-word2vec training.
- Focusing on L_v for the unitary V enhanced the correlation coefficients between the distance vectors of the embedding data of Word2vec and those of Q-word2vec.
- An increase in the number of qubits resulted in a decrease in the correlation coefficients.
- Q-word2vec's performance was slightly less than 70% of Word2vec's performance for the synthetic training data. In addition, we observed that the results of Q-word2vec are relatively affected by the random seed settings.[10]
- For a small corpus, similar groups in Word2vec and Q-word2vec were obtained for word embeddings.
- In the word vector calculations, Q-word2vec's performance was somewhat inferior to Word2vec's performance.

[10] The choice of random seed can lead to instability during training and result in significantly large variances in the outcomes (e.g., Mosbach et al. 2021). Therefore, we consider it desirable to select a final model that demonstrates both high average performance and low variance.

6 Conclusion

In this study, we proposed a quantum circuit called Q-word2vec for word embeddings, which shares a similar structure with the classical Word2vec model. Using Q-word2vec, words can be embedded in a Hilbert space, where the number of words grows exponentially with the system's qubit size. Q-word2vec employs a layered structure consisting of parameterized unitaries and entanglement unitaries, enabling it to learn the relationship between a word and its context.

For training, we proposed a loss function that incorporates the correlation coefficient between the distance vectors of the embedding data and those of the training labels in order to enhance Q-word2vec's performance. Additionally, we presented a heuristic formula for estimating circuit length. The performance of Q-word2vec was evaluated using synthetic data and two small corpora, in comparison with the classical Word2vec model. The results demonstrated promising benefits of Q-word2vec.

Regarding the proposed loss function for Q-word2vec, the observed performance of the model might be constrained by the magnitude of the correlation coefficient term. We note that as the qubit size of Q-word2vec increases, the correlation coefficient tends to decrease. Thus, this could represent a limitation on the model's performance. For training, to achieve a zero error rate, the binary entropy error probability p must be reduced, or the amount of training data (i.e., the number of words) should be decreased.[11] In the experiments, the error probability was set to approximately 0.3 by adjusting the number of embedding qubits (n_e). However, as the system's qubit size (n) increases, achieving $n_e < n$ leads to p asymptotically approaching 0.5, since $n_e \geq (1 - H(p))2^n$. Consequently, the number of words must be reduced relative to n. This limitation represents a drawback of Q-word2vec, as it restricts the effective utilization of the Hilbert space.

To enhance practical applications, it is necessary to address the challenges of applying Q-Word2vec to large-scale datasets and subsequent tasks, such as classification using the embedded data. Additionally, improving the quality of the embedded data may require theoretical advancements in dense quantum coding. In future work, we plan to address this restriction by exploring dense quantum coding and developing quantum generative AI that leverages word embedding data.

[11] For measurements of the output states of V, using excess qubits (as ancilla qubits), the partial measurement technique (Huang et al. 2021) can be employed, thereby enhancing the non-linearity of the model.

Appendix. Correlation between labels in the training data and embedding data obtained from Word2vec

Figure 10 shows the correlation coefficients between the distance vectors of the labels in the training data and those of the embedding data generated by Word2vec. The number of data points was five for each qubit in the training datasets. The error bars represent one standard deviation. From the figure, it can be seen that an increase in the number of qubits leads to a gradual decrease in the correlation coefficients. Thus, for large qubit sizes, a decrease in the loss term value based on the correlation coefficients (Eq. 5) might not result in an increase in the correlation with Word2vec.

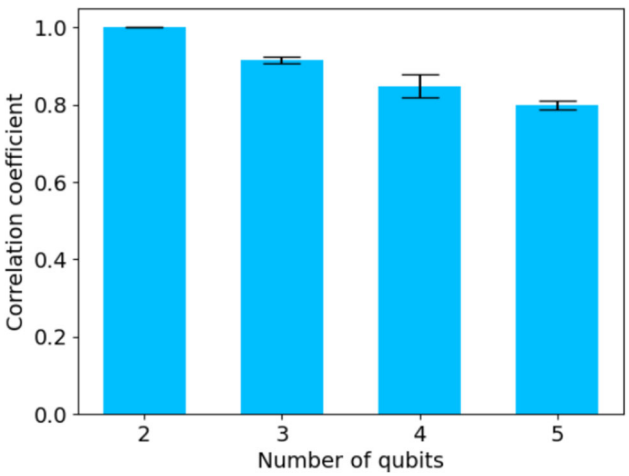


Fig. 10 Results of correlation coefficients between the distance vectors of labels in training data and those of embedding data generated by Word2vec

Author contribution H.O. designed and performed experiments, analysed data and wrote the paper.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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